PART 8

Observability & Reliability Targets

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#058

Monitoring

WHAT IT IS

Continuously collecting health and performance data and watching it.

WHY IT MATTERS

You find problems before customers tell you.

EXAMPLE

CPU, memory, queue depth, DB connections, error rate, request latency on a Grafana/Datadog/Nightwatch dashboard; Laravel Pulse and Horizon for app-level views.

#059

Logging

WHAT IT IS

Recording events the app emits.

WHY IT MATTERS

Primary tool for debugging what happened.

EXAMPLE

```
Log::withContext(['request_id' => $requestId,
    'user_id' => auth()->id()]);
Log::warning('payment.failed', ['order' =>
    $order->id, 'reason' => $e->getMessage()]);
```

#060

Distributed Tracing

WHAT IT IS

Following one request across many services as a tree of timed "spans".

WHY IT MATTERS

Pinpoints which service or query made the request slow.

EXAMPLE

```
Http::withHeaders(['traceparent' => request(
    )->header('traceparent')])->get($billingUrl);
```

#061

Metrics

WHAT IT IS

Numeric time-series (counters, gauges, histograms).

WHY IT MATTERS

Cheap to store, great for dashboards and alerts.

EXAMPLE

```
http_requests_total{route="/tasks",status="500"} 12
http_request_duration_seconds_bucket{le="0.5"} 9812
queue_depth{queue="emails"} 340
```

#062

Alerts

WHAT IT IS

Automatic notifications when a metric crosses a threshold.

WHY IT MATTERS

Humans cannot stare at dashboards; bad alerts cause fatigue.

EXAMPLE

```
- alert: HighErrorRate
  expr: sum(rate(http_requests_total{status=~"5.."}[5
    m])) / sum(rate(http_requests_total[5m])) > 0.02
  for: 10m
  labels: { severity: page }
```

#063

SLOs

WHAT IT IS

Service Level Objectives: the reliability target you promise internally.

WHY IT MATTERS

Turns "be reliable" into a measurable goal that guides priorities.

EXAMPLE

"99.9% of API requests succeed and 95% finish under 300ms over 30 days."

#064

SLIs

WHAT IT IS

Service Level Indicators: the actual measurements that SLOs are based on.

WHY IT MATTERS

You cannot hit a target you do not measure.

EXAMPLE

SLI = successful_requests / total_requests; latency SLI = share of requests faster than 300ms.

#065

Error Budgets

WHAT IT IS

The allowed unreliability: 100% - SLO.

WHY IT MATTERS

Balances shipping speed and stability. Budget left → ship faster; budget burned → freeze features, fix reliability.

EXAMPLE

99.9% over 30 days = ~43 minutes of allowed downtime. Incident used 30 minutes → only ~13 left this month.

#066

Observability

WHAT IT IS

Ability to understand internal state from outputs (logs, metrics, traces) and ask new questions without shipping new code.

WHY IT MATTERS

Monitoring tells you that something broke; observability helps you find why.

EXAMPLE

A spike in 500s → trace shows /export slow → span shows a missing index on tasks.project_id → log line carries the offending tenant ID. Correlate all three with a shared request_id.

Found this useful?

Save it for later.

Share it with a teammate.

Full guide with all code examples:
the Backend Engineering Handbook (GitHub).